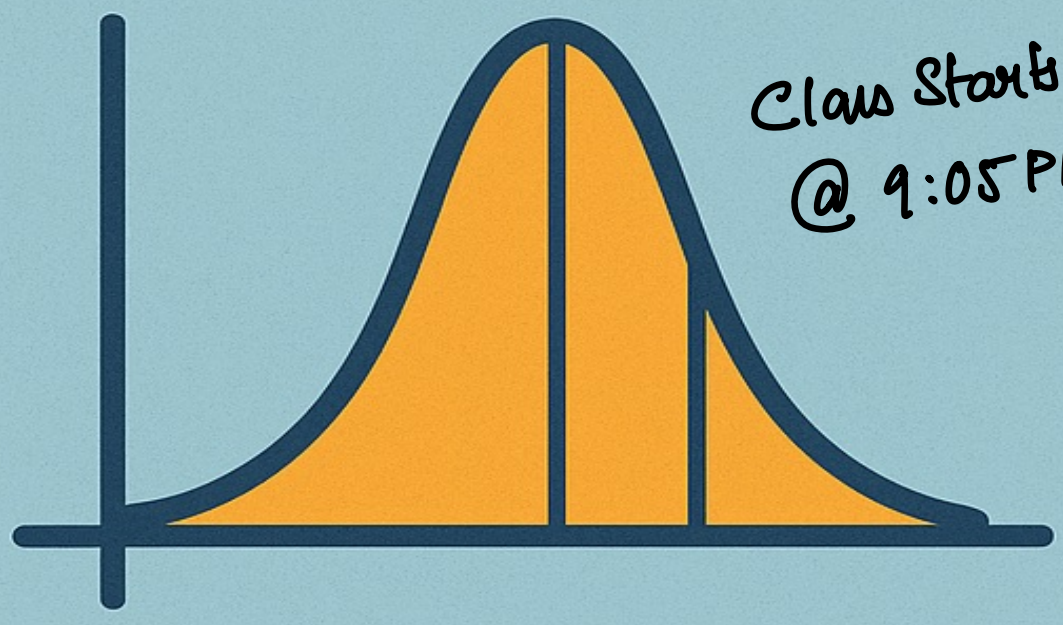


PROBABILITY AND STATISTICS





Lecture 8: Probability Distribution 3

Class starts at 9:05 PM

#Agenda

1. Recap – Uniform distribution ✓
2. Gaussian / Normal Distribution ✓ → Main topic most popular
3. 68/95/99 rule (Emperical Rule)
4. Z-Score
5. Percent Point function (PPF)
6. Standard Normal distribution

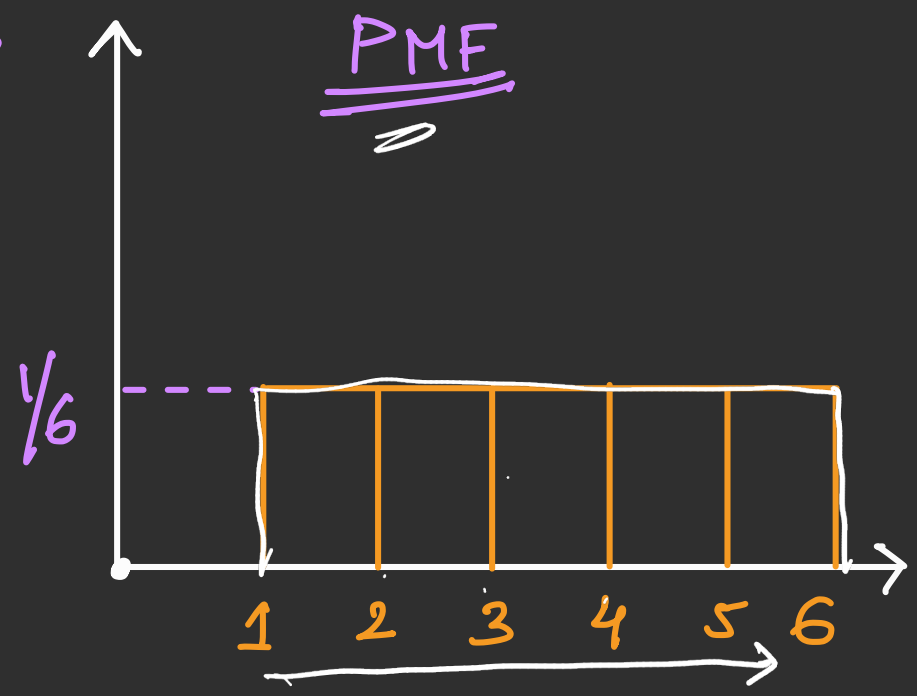
→ Normal Distribution

Uniform Distribution

P(every outcome is same)

① Dice → $P(X=1) = 1/6$, $P(X=2) = 1/6$, ⋮, $P(X=6) = 1/6$

probability → all event is equaly likely



② Cards : $P(\text{Heart}) = P(\text{Spade}) = P(\text{Diamond}) = P(\text{Club})$

13 → 13 → (1/13)

13 → 13

13 → 13



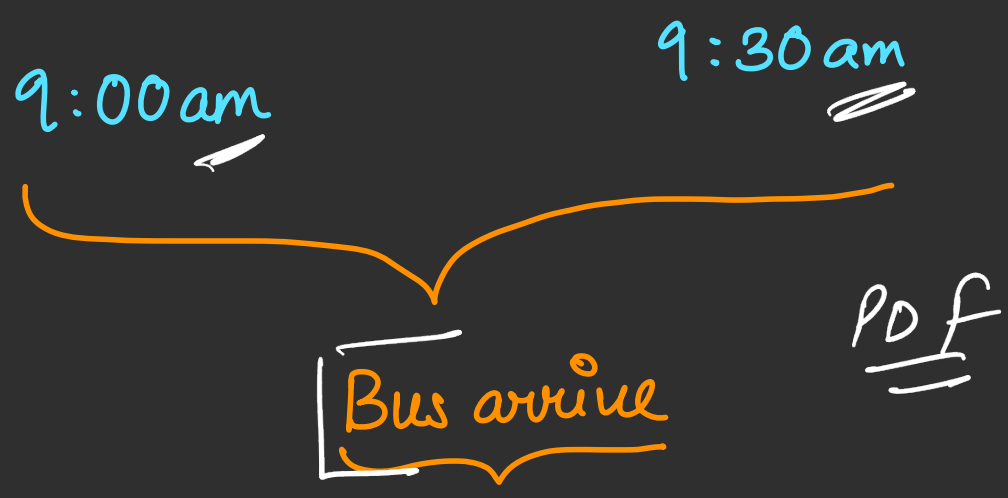
0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0

$P(X=0.1) = P(X=0.5) = P(X=0.7) = P(X=0.9)$

$0 \longrightarrow 1$

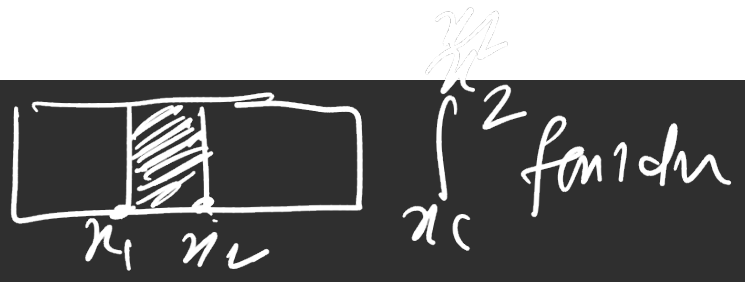
$P(\infty) = ?$

↳ infinite



Uniform dist for cont RV

Discrete Uniform Distribution (PMF)



Uniform dist is applicable for both Const & Dist RV.

[1] 2, 3, 4, 5, [6]

$a = 1 \quad b = 6$

a & b [3, 5]

$$PMF(X) = \frac{1}{(b-a+1)}$$

where $a \le x \le b$

3, 4, 5

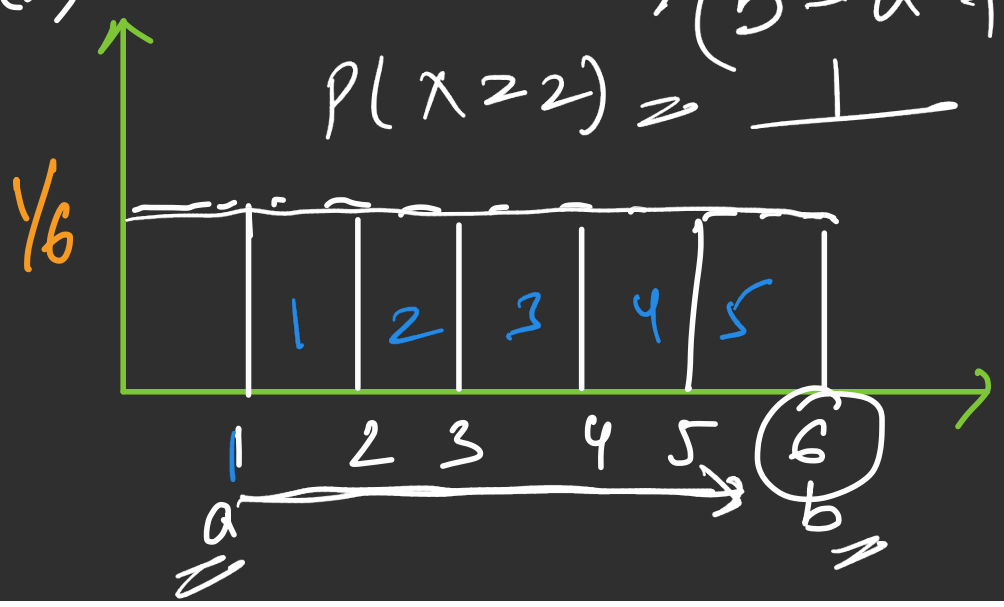
$a \rightarrow$ lower limit

$b \rightarrow$ upper limit

$$= \frac{1}{(b-a+1)}$$

$f(x)$

$P(x=2) = \frac{1}{1}$



$P(x=1) = (1/6)$

$P(x=1) = \frac{1}{6-1+1} = (1/6)$

If $a = 3, b = 5$

the no.'s are $3, 4, 5 \rightarrow$ 3 numbers

$b - a = 5 - 3 = 2$

$= 2 + 1$

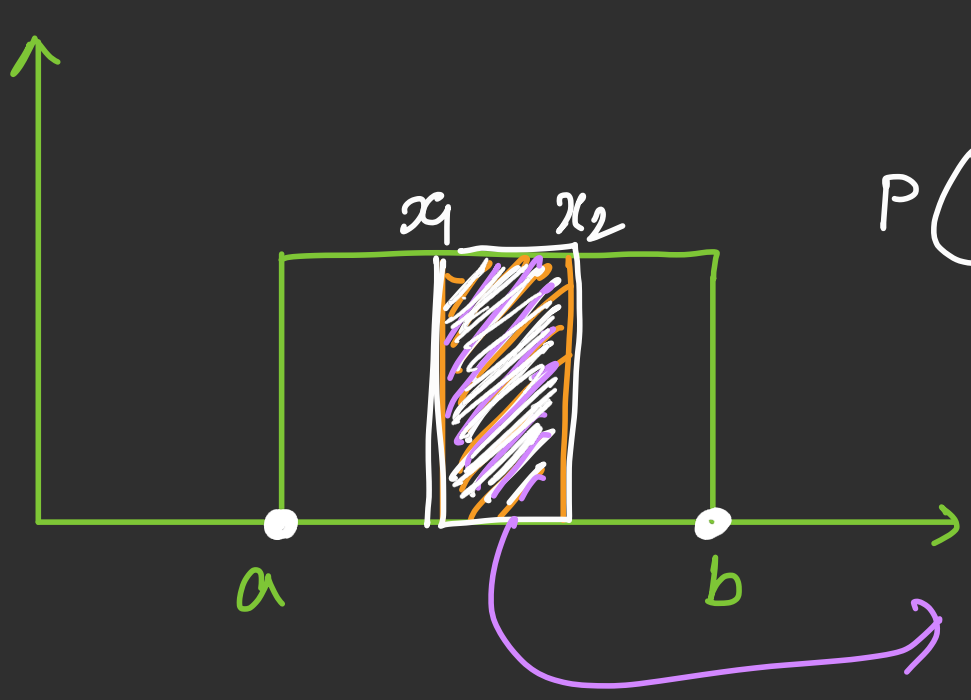
$= 3$

Continuous Uniform Distribution (PDF)

$$PDF(x) = \frac{1}{(b-a)}$$ where $a \leq x \leq b$

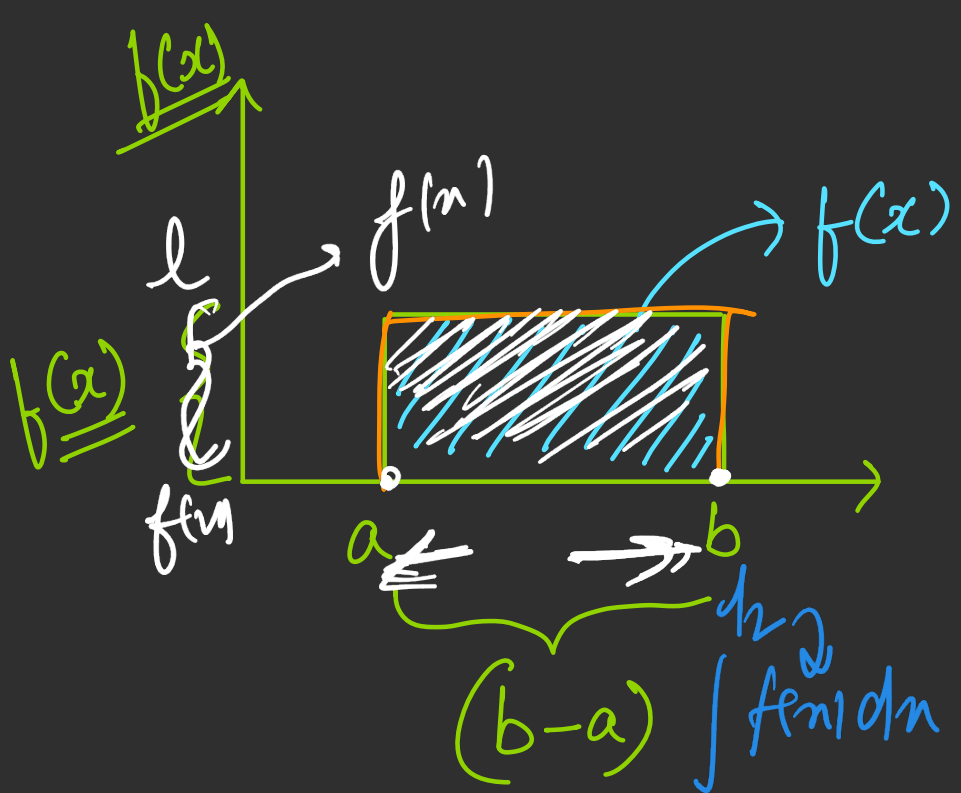
$a \rightarrow$ lower limit

$b \rightarrow$ upper limit



$$P(x_1 \leq X \leq x_2) = \frac{x_2 - x_1}{(b-a)}$$

Integration $= \int_a^b f(x)\,dx$



$$= l \times b$$
$$= a \times a$$

$$\text{area} = f(x) \times (b-a)$$

$$1 = f(x) \times (b-a)$$

$$f(x) = \frac{1}{(b-a)}$$

$$(b-a) \int_{l_1}^{l_2} f(x)\, dx$$

$$a = l \times b \quad ,$$

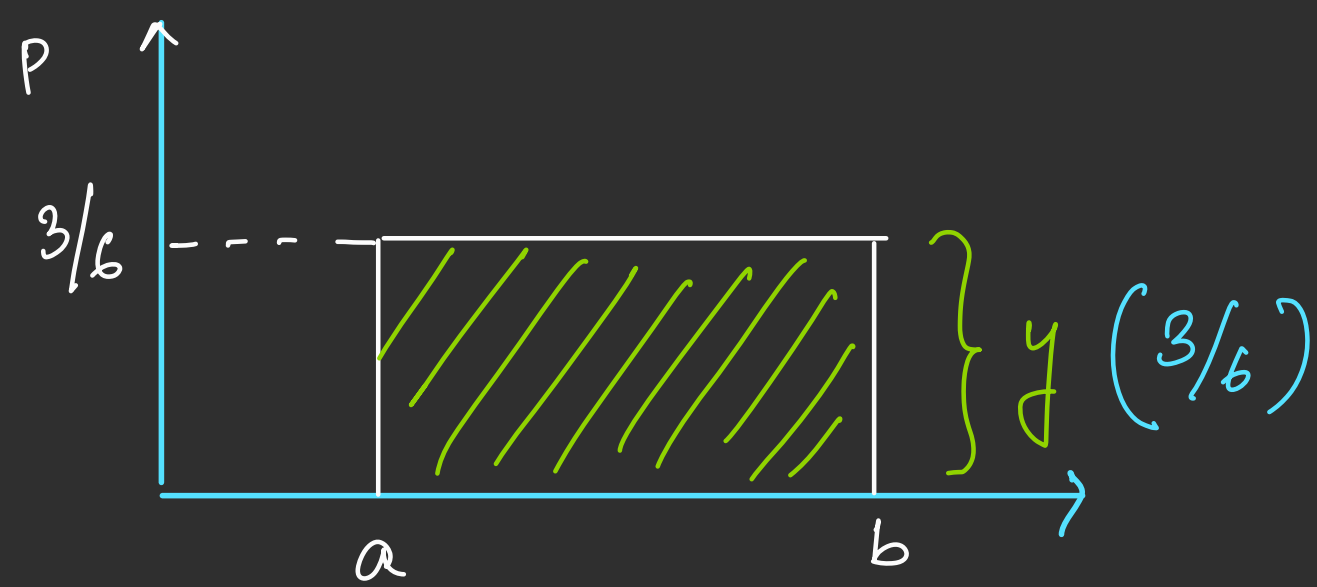
$$PDf = \frac{1}{b-a}$$

$$\text{area} = l \times (b-a)$$
$$1 = l \times (b-a)$$
$$1 = \frac{1}{b-a}$$

Apply uniform dist

```
from scipy.stats import uniform
```

- uniform.pmf
- uniform.pdf
- uniform.cdf



$$\underbrace{\qquad\qquad}_{x}$$
$$(b-a)$$

$$\underline{\underline{ar}} = (b-a) \times 3/6$$

Statistics Related to Uniform Dist

1. Mean = $(a+b)/2$

a & b are limit

2. Variance for cont RV = $\frac{(b-a)^2}{12}$

3. Variance for disrk RV = $\frac{(b-a+1)^2 - 1}{12}$

4. std = $\sqrt{var}$

$q = (1-p)$

Binomial
Mean = np
Variance = npq
$= npq$

Binomial dist

$E(x) = n \times p$

- n: No of trial
- p: Prob Succes

$Var = np(1-p)$

Gaussian / Normal Distribution [Continous RV]

(same)

Height → class

↳ Avg (5.5 to 5.7)

100 — learners

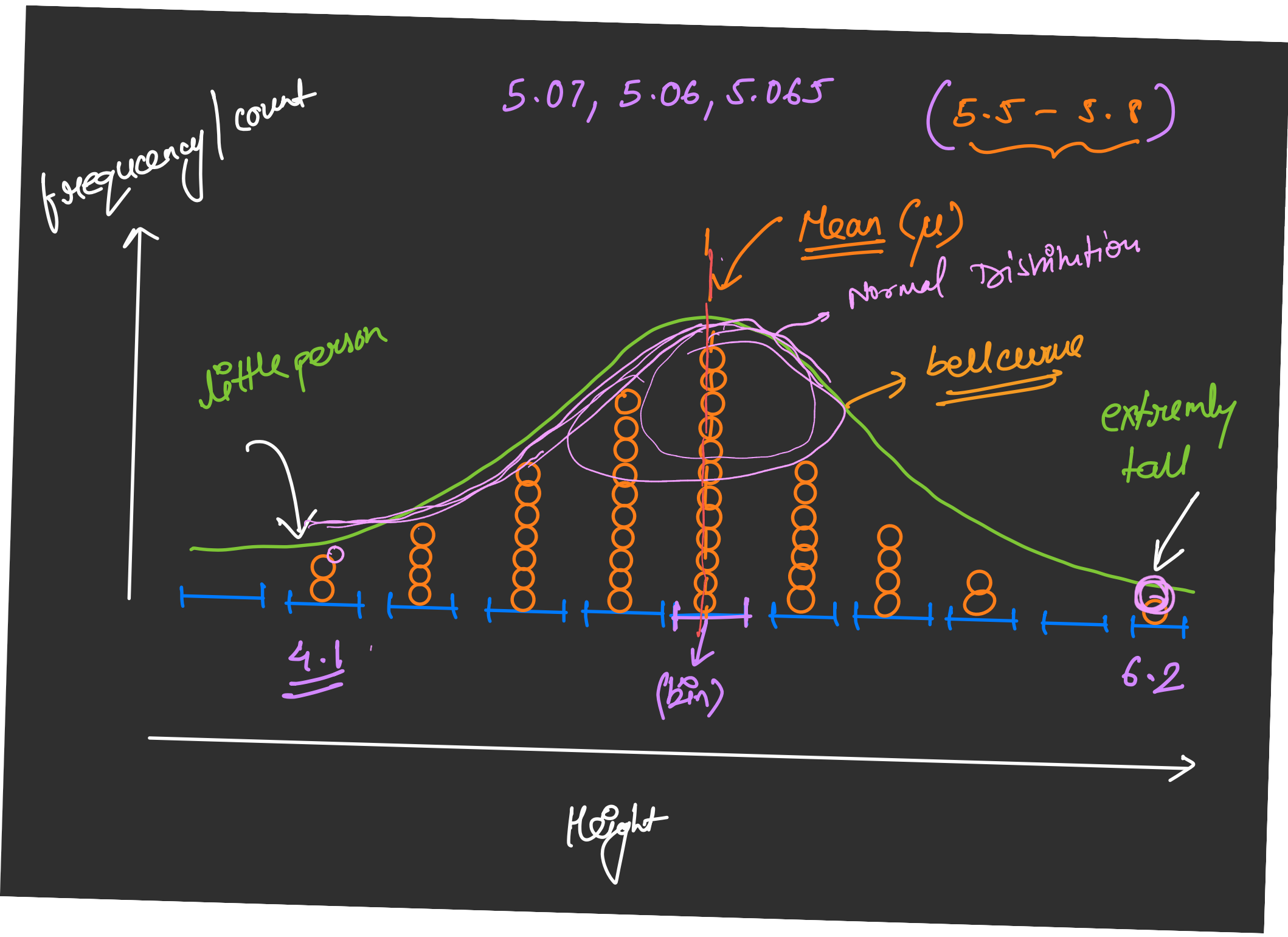
100 → data point

{pd.cut()}

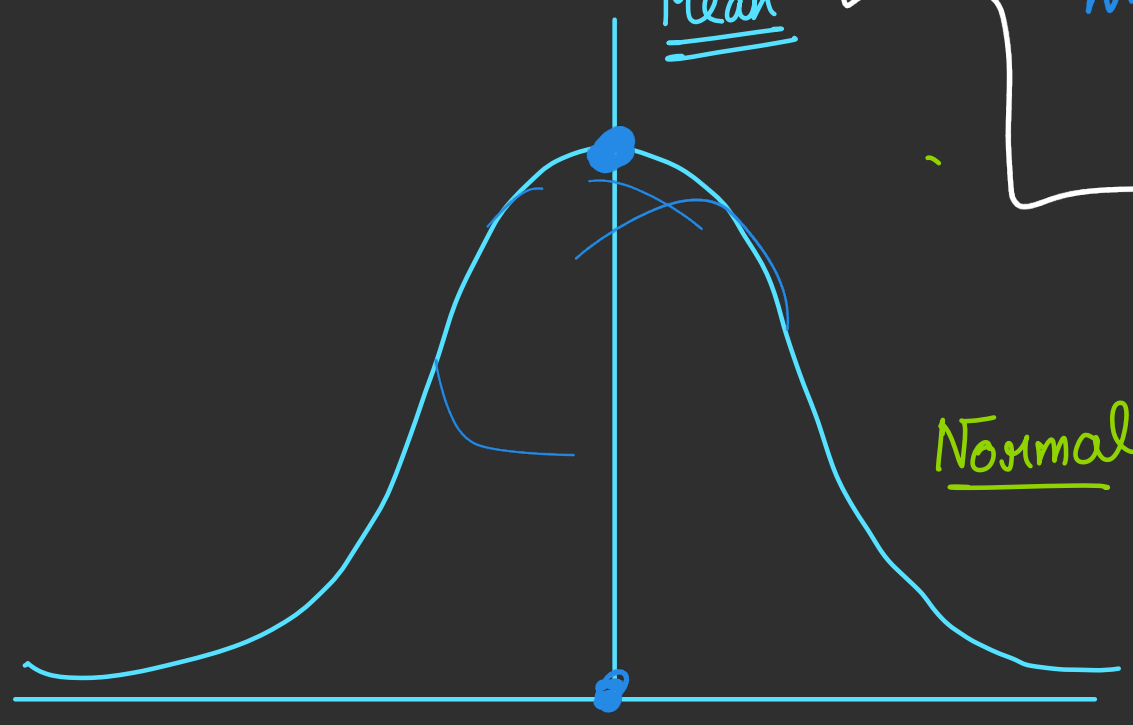
If I want to get avg height of class is this achievable ?,

30 / 5.1–5.2 = 5.125

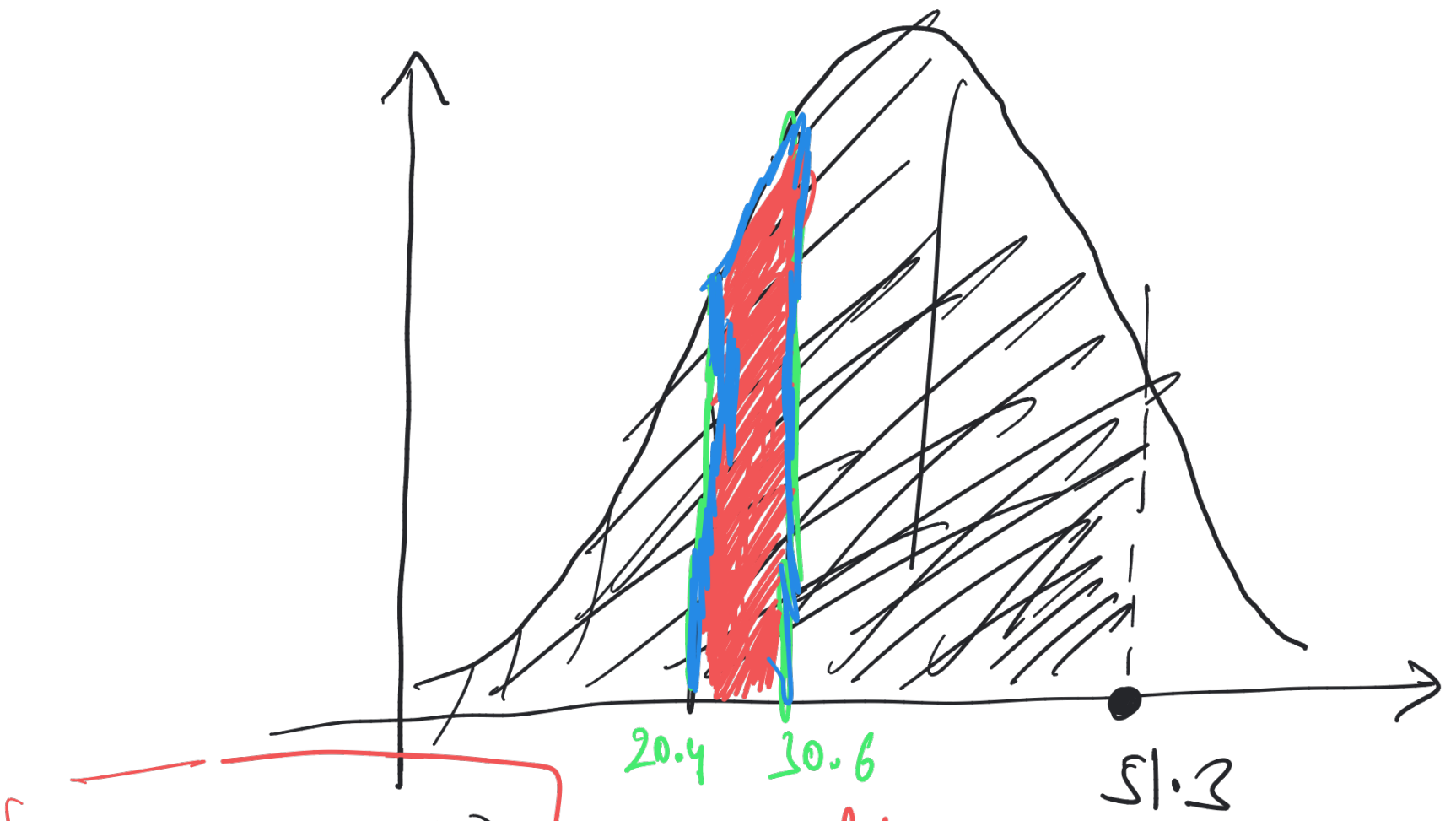
6 / 5.21–5.3 = 5.296


frequency / count
5.07, 5.06, 5.065
(5.5 - 5.8)
Mean (μ)
Normal Distribution
little person
bell curve
extremely tall
4.1
(bin)
6.2
Height

In normal distribution
mean = median
= Mode



Normal dist?



$P(X \leq 51.3)$ → norm.cdf

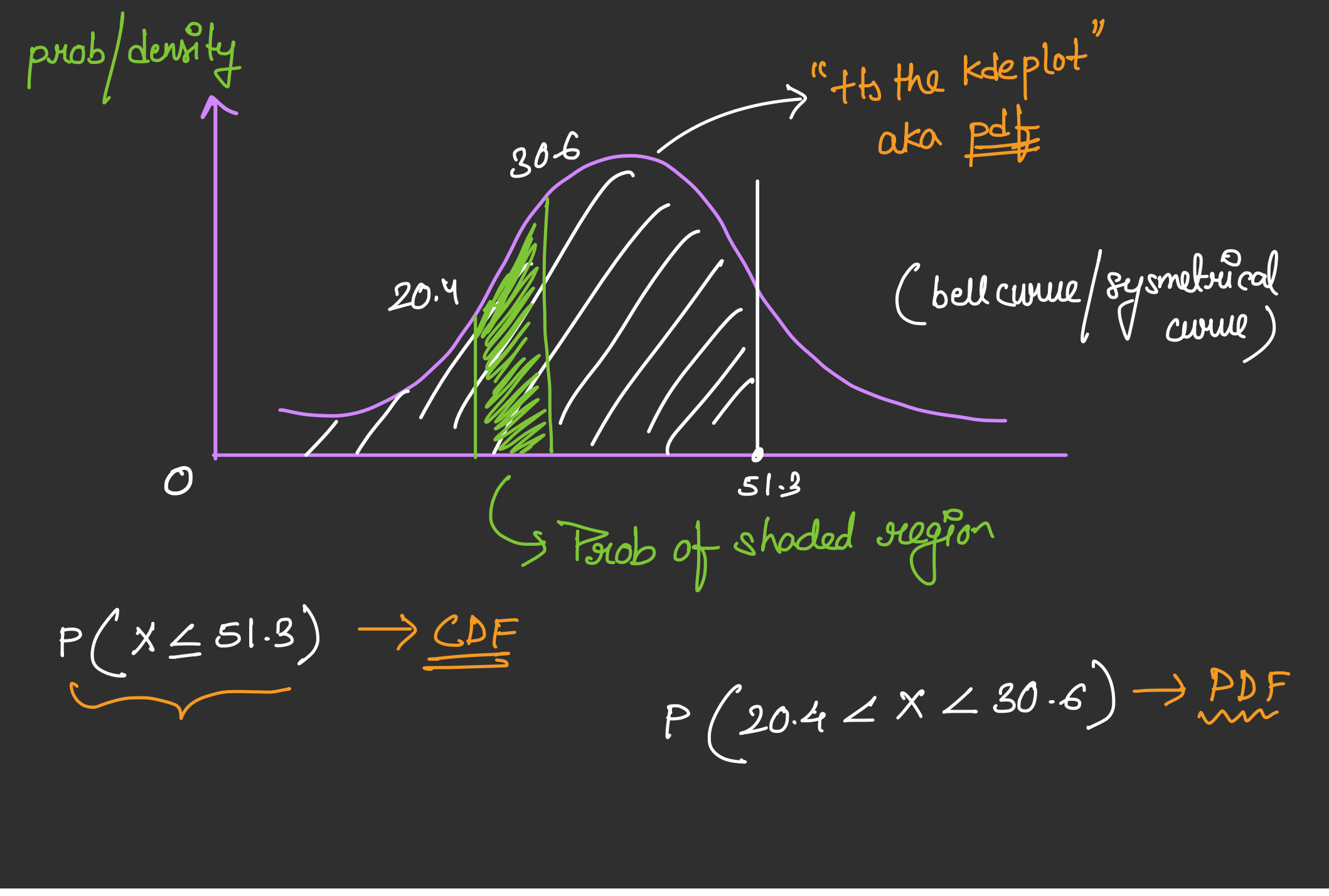
norm.cdf

$P(X \geq 51.1) = 1 - \text{norm.cdf}($

$P(20.4 < X < 30.6)$ → PDF

normal.pdf

norm.pdf


prob/density
"this the kdeplot"
aka pdf
30.6
20.4
(bell curve/symetrical curve)
0
51.3
Prob of shaded region
P(X ≤ 51.3) → CDF
P(20.4 < X < 30.6) → PDF

Popular Interview Question

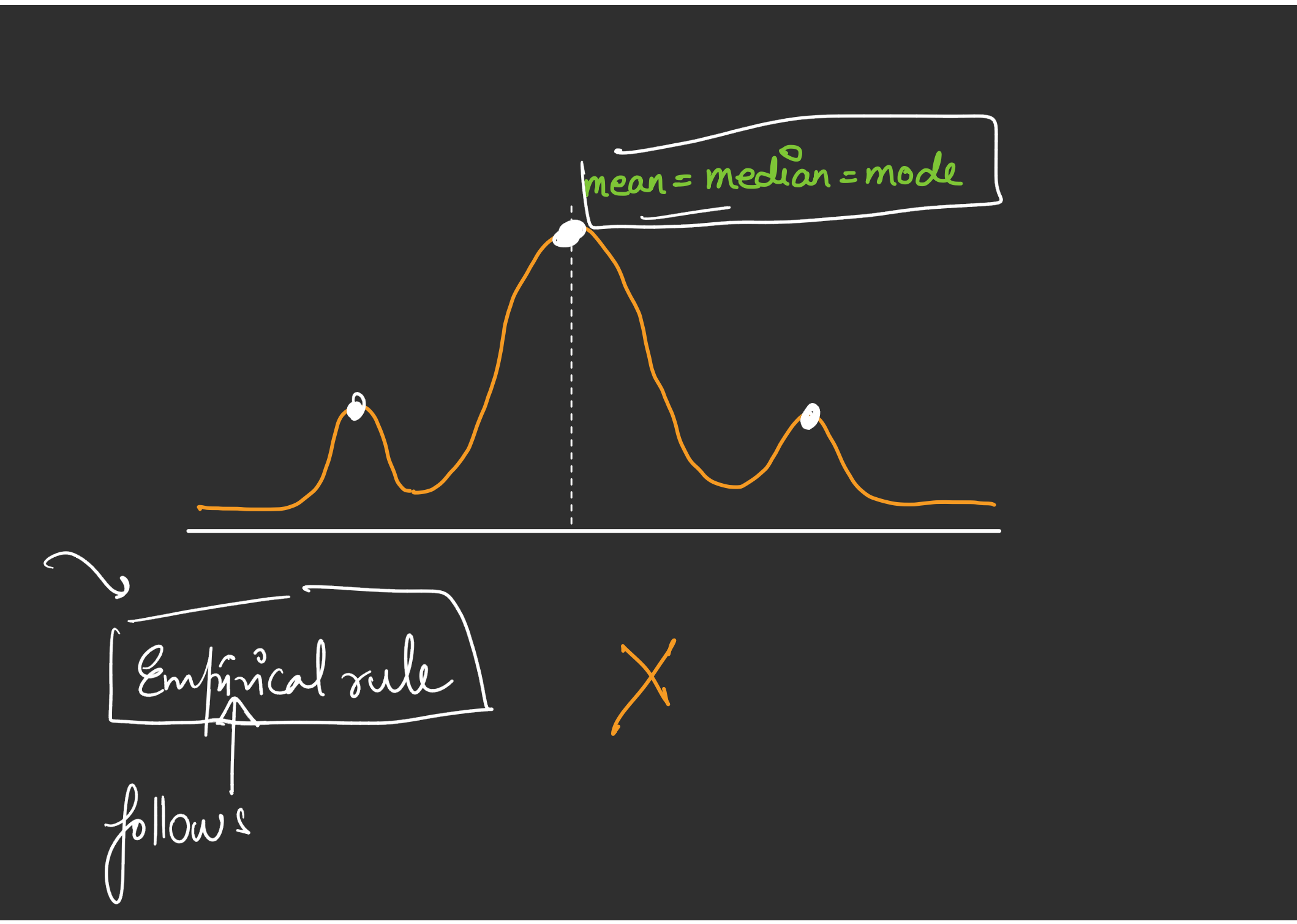
Mean = Mode = Median

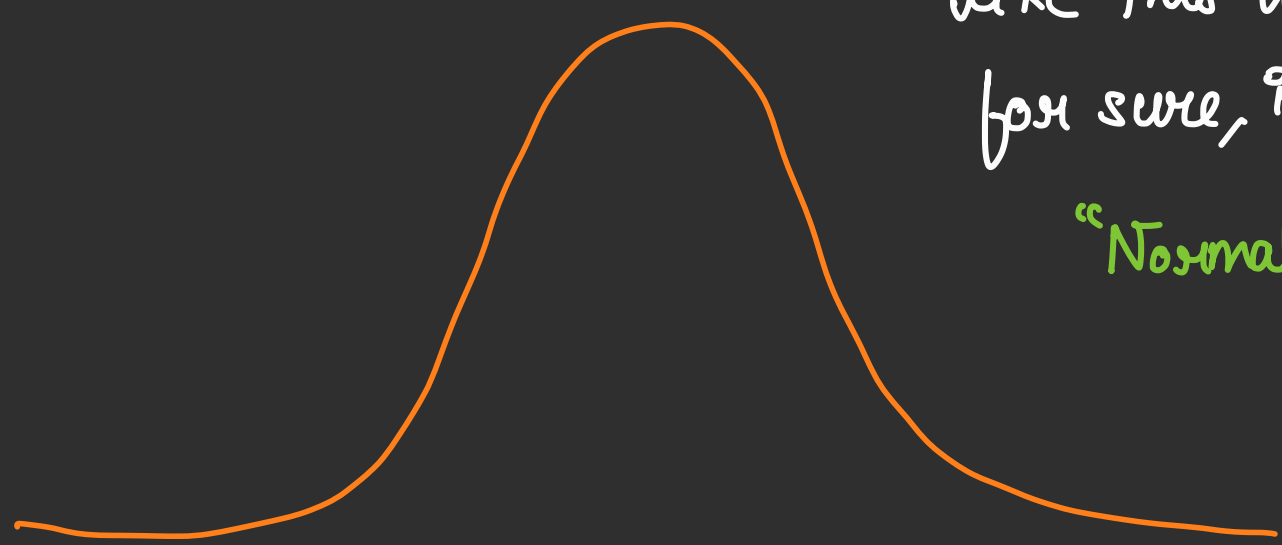
50% percentile



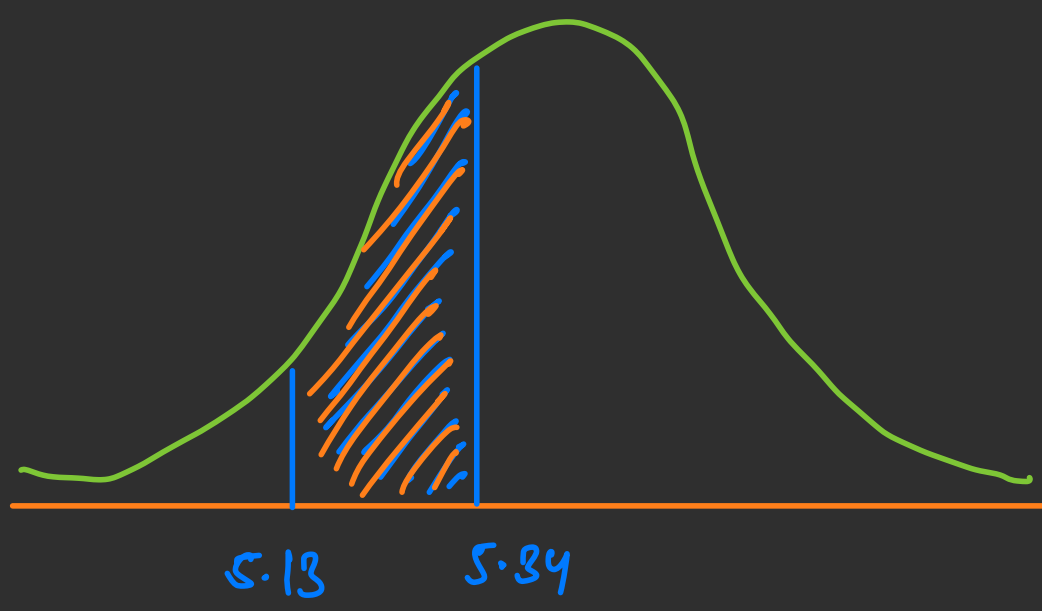
Specific property ND ⇒ Mean = Median = Mode

Not Necessarily


mean = median = mode
Empirical rule
follows
X



Even if the curve looks like this we cannot tell for sure, if it's "Normal Dist"


5.13
5.34

Empirical Rule [68/95/99 Rule]

↳ Experimental

$$PDF = \frac{1}{b-a}$$

data

↳ mean = 65 inches

SD = 2.5 inches

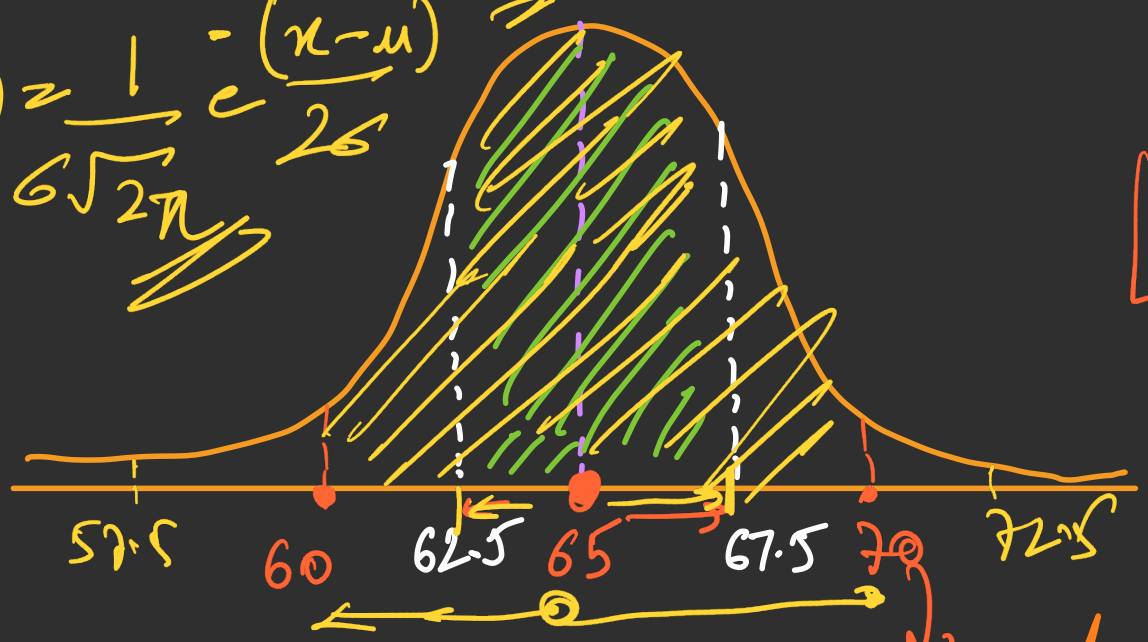
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma}}$$

68%

$\mu = 65$

67.5 2 SD

62.5



$65 + 1(2.5) = 67.5$

$65 - 1(2.5) = 62.5$

$65 + 2 \times 2.5 = 70$

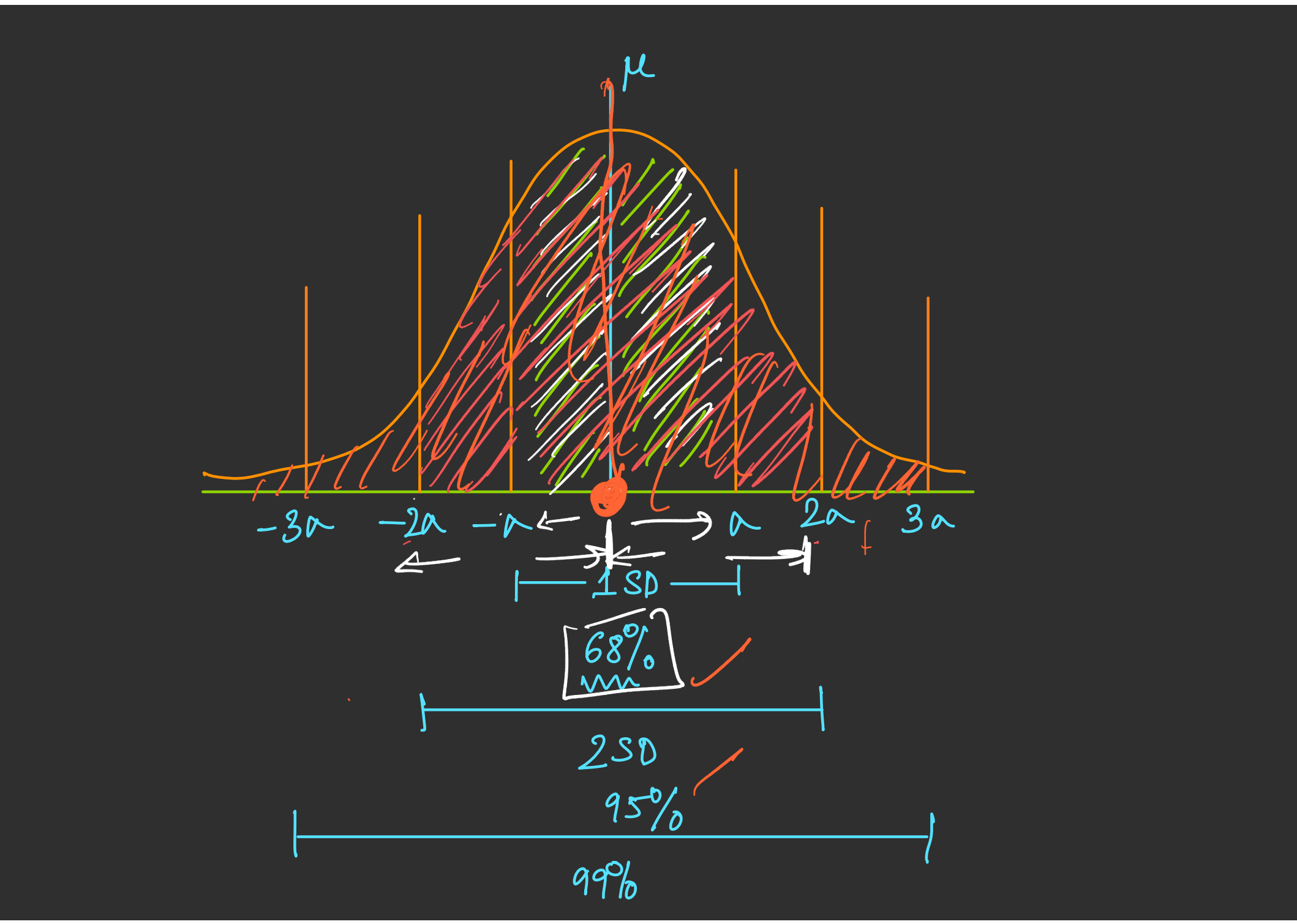
$65 - 2 \times 2.5 = 60$

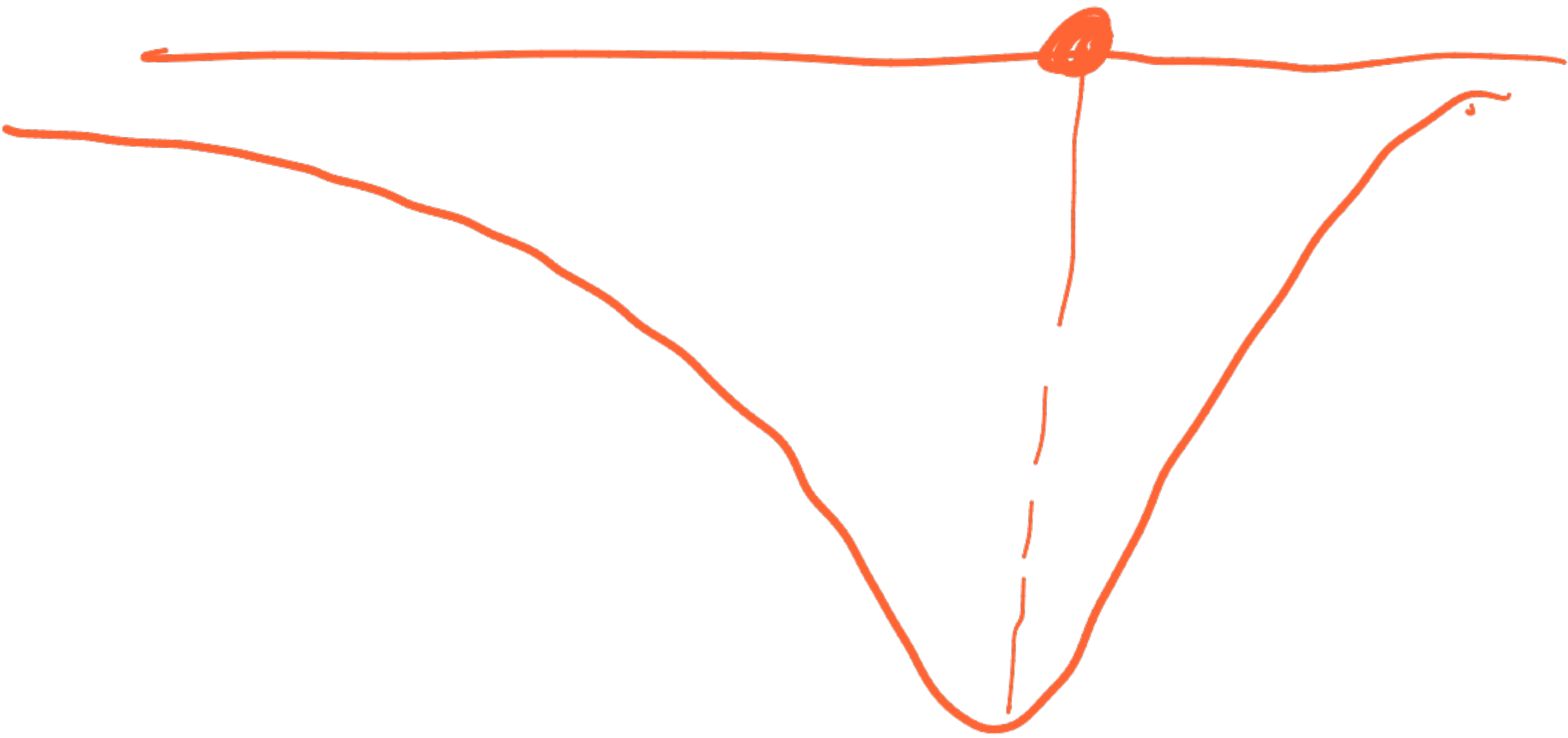
$65 - 3 \times 2.5 = 57.5$

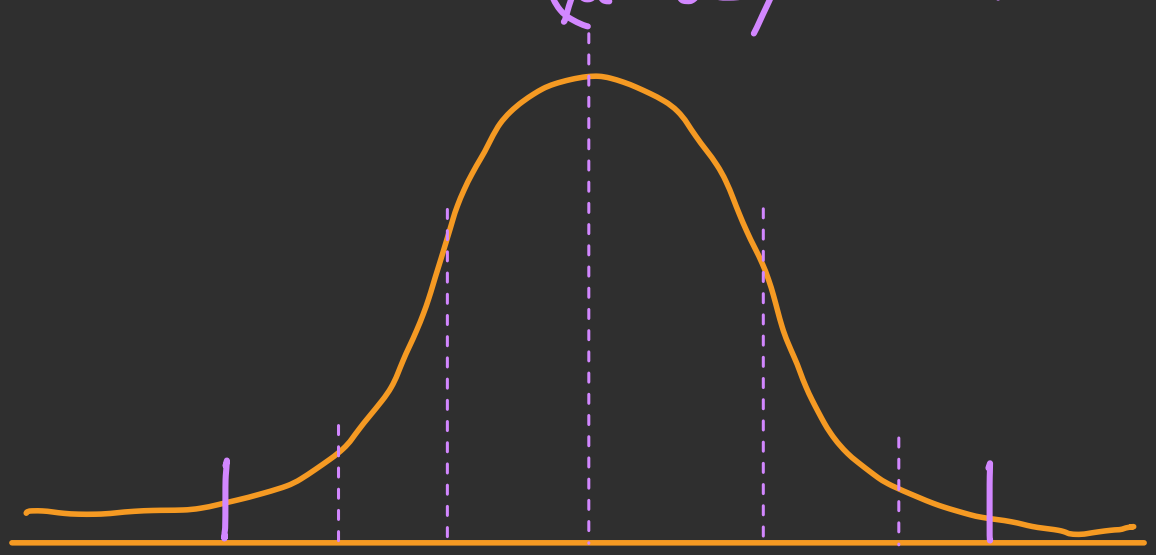
$65 + 3 \times 2.5 = 72.5$

$P(62.5 < X < 67.5) \Rightarrow 0.68 / 68\%$

$P(60 < X < 70) \Rightarrow 95\%$


μ
-3σ
-2σ
-σ
σ
2σ
3σ
1 SD
68%
2SD
95%
99%





Case I

$65 + 2.5 = 67.5$

$65 - 2.5 = 62.5$

Case II

$65 + 2(2.5) = 70$

$65 - 2(2.5) = 60$

Case III

$65 + 3(2.5) = 72.5$

$65 - 3(2.5) = 57.5$

$P(62.5 < X < 67.5) = 0.68$ / 68%

$P(60 < X < 70) = 0.95$ / 95%

$P(57.5 < X < 72.5) = 0.99$ / 99%

$P(62.5 < X < 67.5) = 0.68$ / 68%

$P(60 < X < 70) = 0.95$ / 95%

$P(57.5 < X < 72.5) = 0.99$ / 99%

↳ Normal Dist

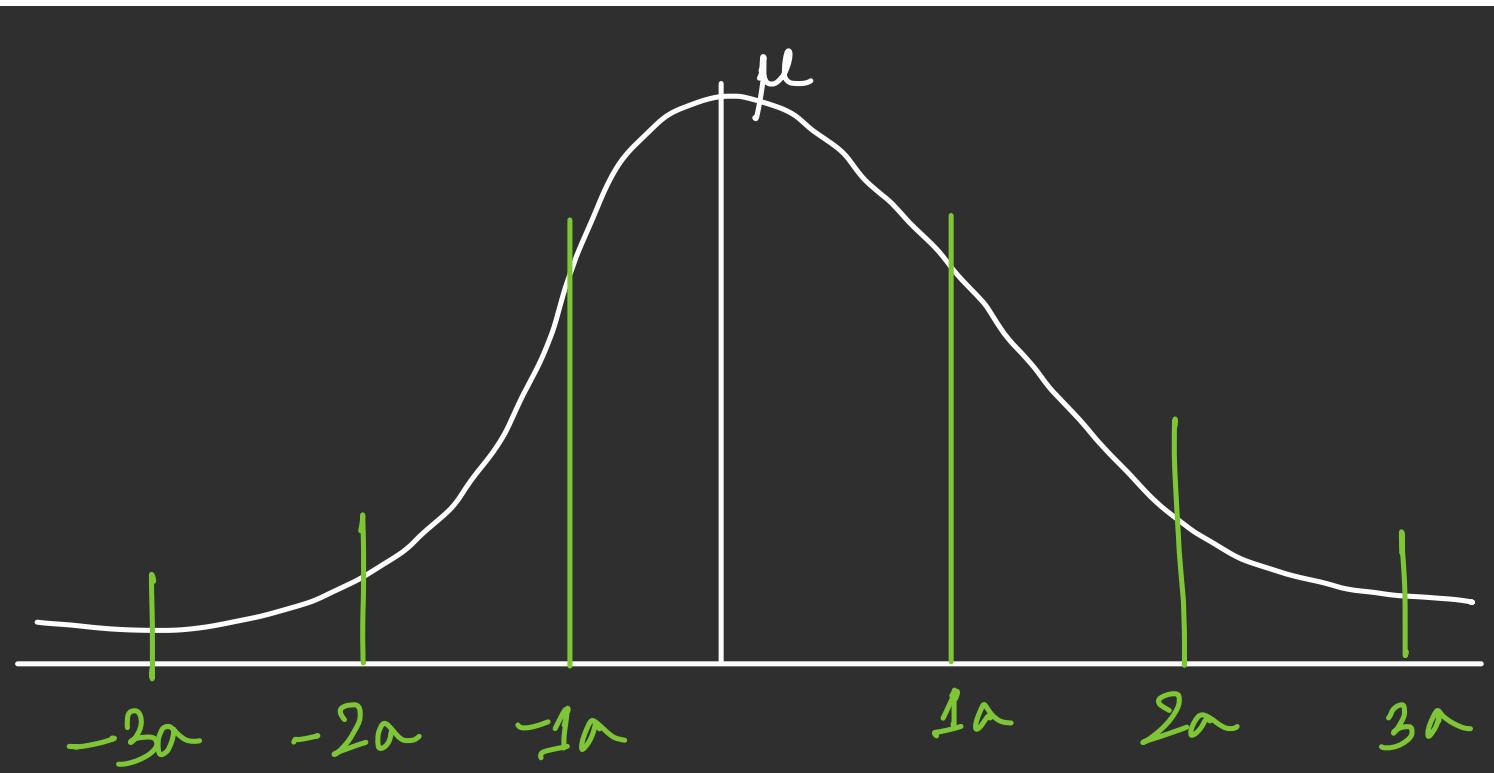
$65 + 2(25)$

$[\mu + 2\sigma, \mu - 2\sigma] \longrightarrow$ 95%

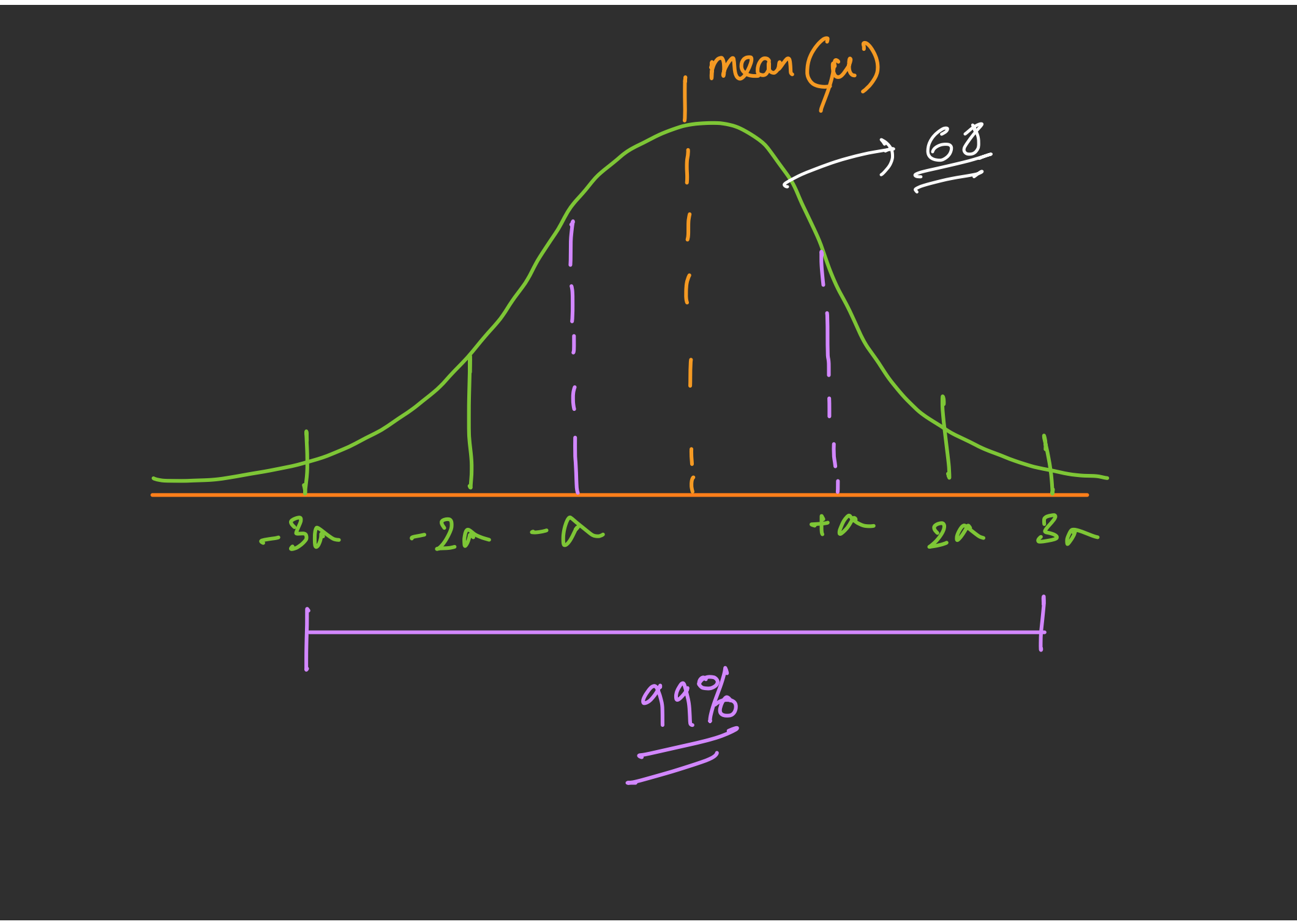
$[\mu + \sigma, \mu - \sigma] \longrightarrow$ 68%

$[\mu + 3\sigma, \mu - 3\sigma] \longrightarrow$ 99%

If your data is normal, then this rule applies



99% → $[\mu \pm 3\sigma]$

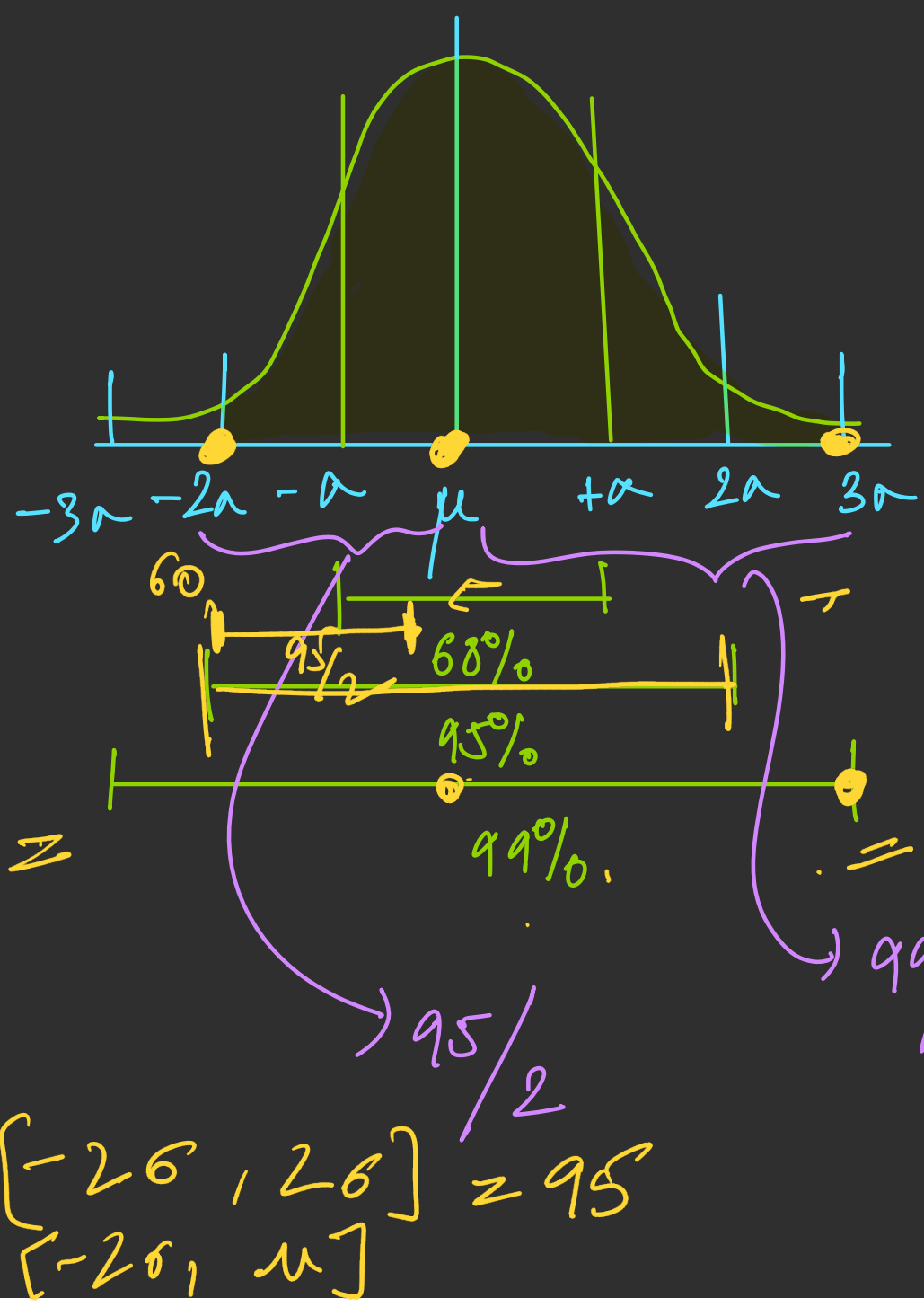

mean (μ)
6σ
-3σ
-2σ
-σ
+σ
2σ
3σ
99%

PDF formula for Normal dist

Use library

No need to remember this

$$pdf = \frac{1}{\sigma\sqrt{2\pi}} e^{\left\{\frac{(x-\mu)^2}{-2\sigma}\right\}}$$



$P(60 < X < 72.5) = ?$

$65 - 2 \times 2.5$

$= 60$

$\mu = 65$

$\sigma = 2.5$

$65 + 3 \times 2.5$

$= 72.5$

$(-2\sigma \text{ to } \mu \text{ to } 3\sigma)$

$[-2\sigma, 2\sigma] = 95$

$[-2\sigma, \mu]$

$\Rightarrow \frac{95}{2} + \frac{99}{2}$

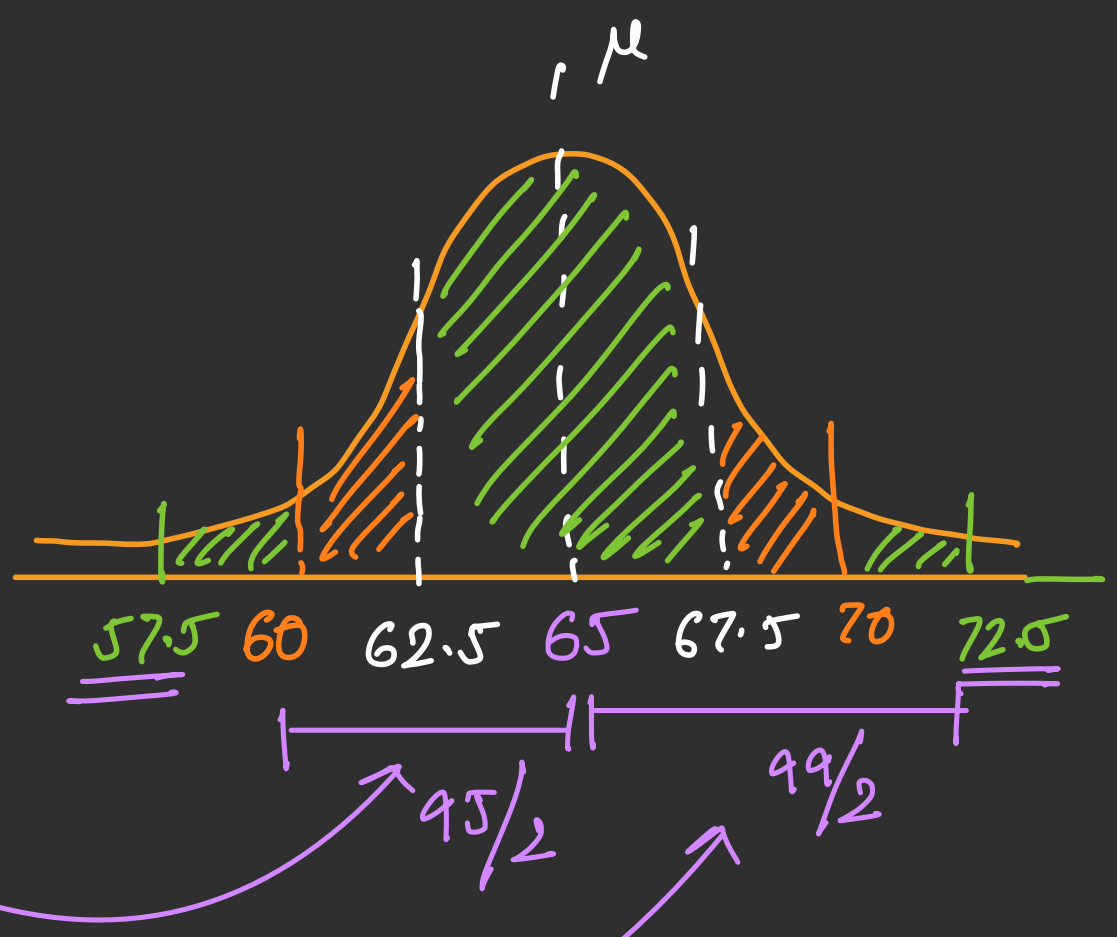
97.35%

#Quiz 1

Data is Normal

$$\begin{cases} \mu = 65 \\ \sigma = 2.5 \end{cases}$$

$P(60 < X < 72.5)$



$P(60 < X < 70) = 95$

$P(57.5 < X < 72.5) = 99$

$95/2 + 99/2 \Rightarrow 0.9735$

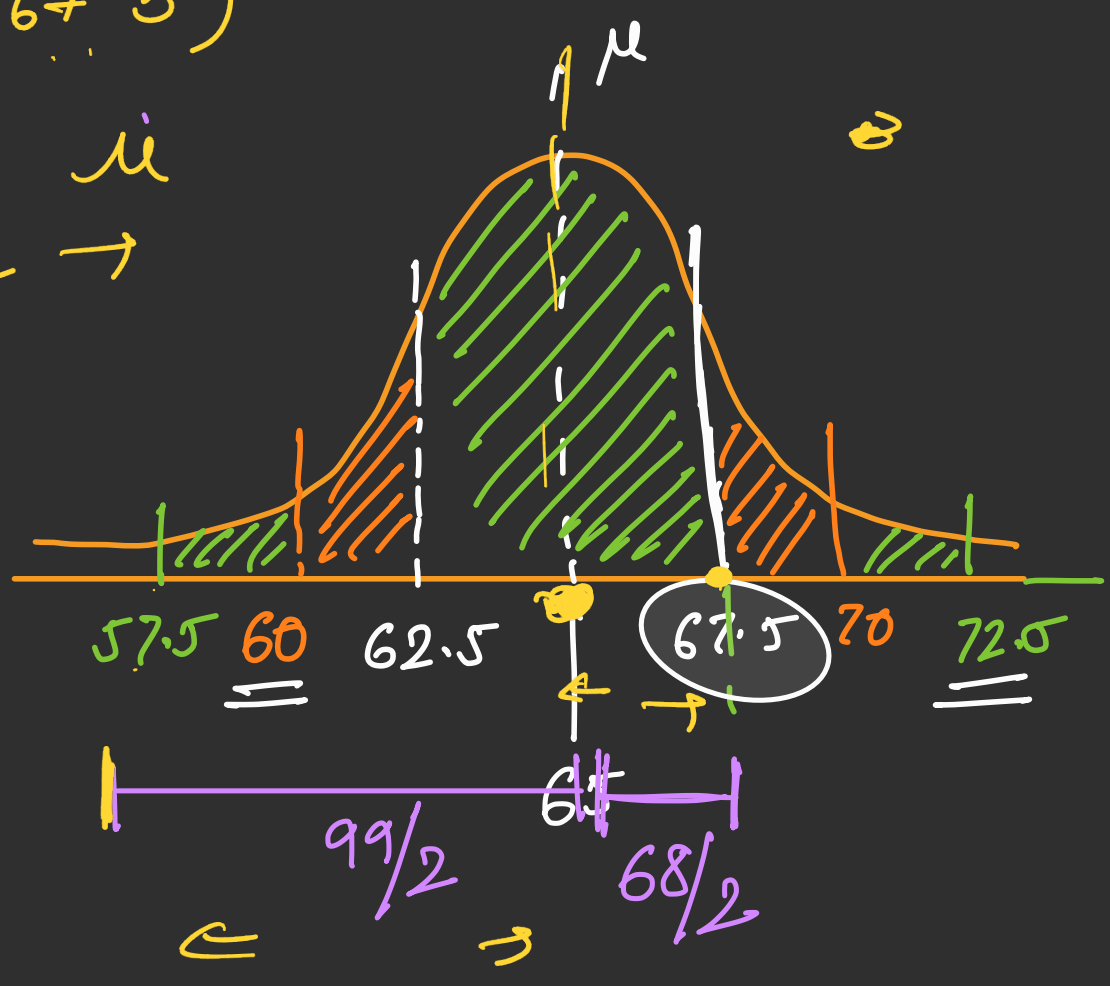
#Quiz 2

$P(X \leq 67.5)$

$\begin{cases} \mu = 65 \\ \sigma = 2.5 \end{cases}$

$-3\sigma \quad \mu$

$\leftarrow \sigma \rightarrow$



$P(X \leq 67.5) = ?$

$[-3\sigma, \mu] \rightarrow 99/2$

$[\mu, 1\sigma] \rightarrow 68/2$

$P(X \leq 67.5) = 99/2 + 68/2 \Rightarrow 0.84$

Understanding Z-score

$P(X < 67.5) \rightarrow$

$P(X < 69.1) \rightarrow$

$P(X < 69.1) = 95\%$



$65 + (1)(2.5) = 67.5$

$65 + (2)(2.5) = 70$

$\mu = 65$

$\sigma = 2.5$

$P(X < 72.5) \rightarrow 96$

B/w 1 SD & 2SD

What fraction of people are shorter than 69.1?

$65 + z(2.5) = 69.1$

$z = \frac{69.1 - 65}{2.5}$

Is 69.1 falling on any of the 2σ's

69.1

70 inches → 2 SD from mean

mean → $65 + 2(2.5)$

67.5 inches is 1 SD from mean

$65 + 1(2.5)$

$P(X < 67.5)$

$65 + 1 \times 2.5$

$= 67.5$

$P(X < 67.5)$

$P(X < 69.1)$

67.5 70

$69.1 = 65 + z(2.5)$

$z = \frac{69.1 - 65}{2.5}$

normalcdf(1.64)

$\Rightarrow 1.64$

normalcdf(2)

$$65 + Z(2.5) = 69.1$$

$$Z = \frac{69.1 - 65}{2.5}$$

w/ $(Z = 1.64)$

$$z\ \text{score} = \frac{x - \mu}{\sigma}$$

where x is the pt you are trying to find the z value for

z score value $\Rightarrow$ $\underline{1.64}$ ✓

$\underline{x = 69.1}$

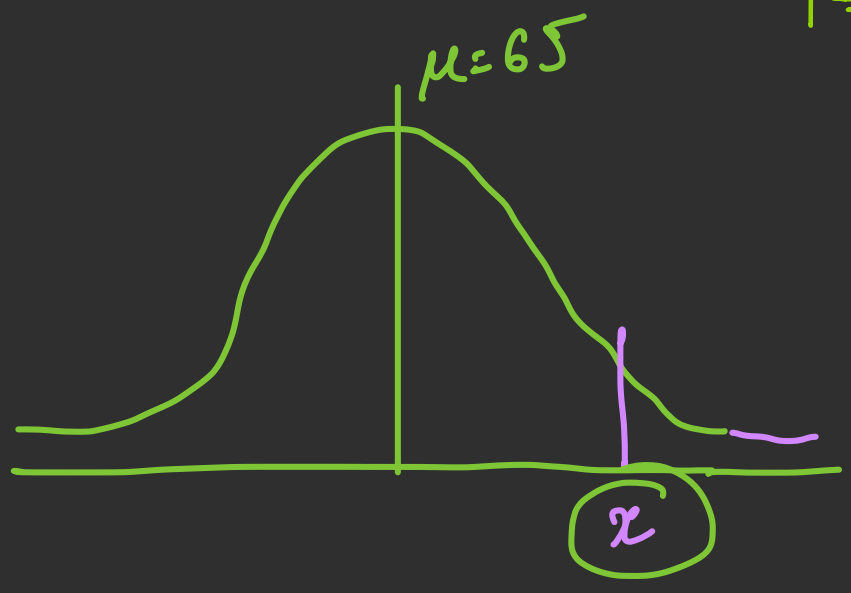
$$P(X < 69.1) = ??$$

$$\text{Z score} = 1.64$$

PPF (Percent Point Function) → percentage or proportion

The height of people is Gaussian with mean 65 inches and standard deviation 2.5 inches. One person says, 96% people are shorter than me. What is my height?

↳ percentile



percentile → zscore

zscore → probability

percentile ⟶ z score

norm.ppf(0.96)

ppf is the opp of zscore

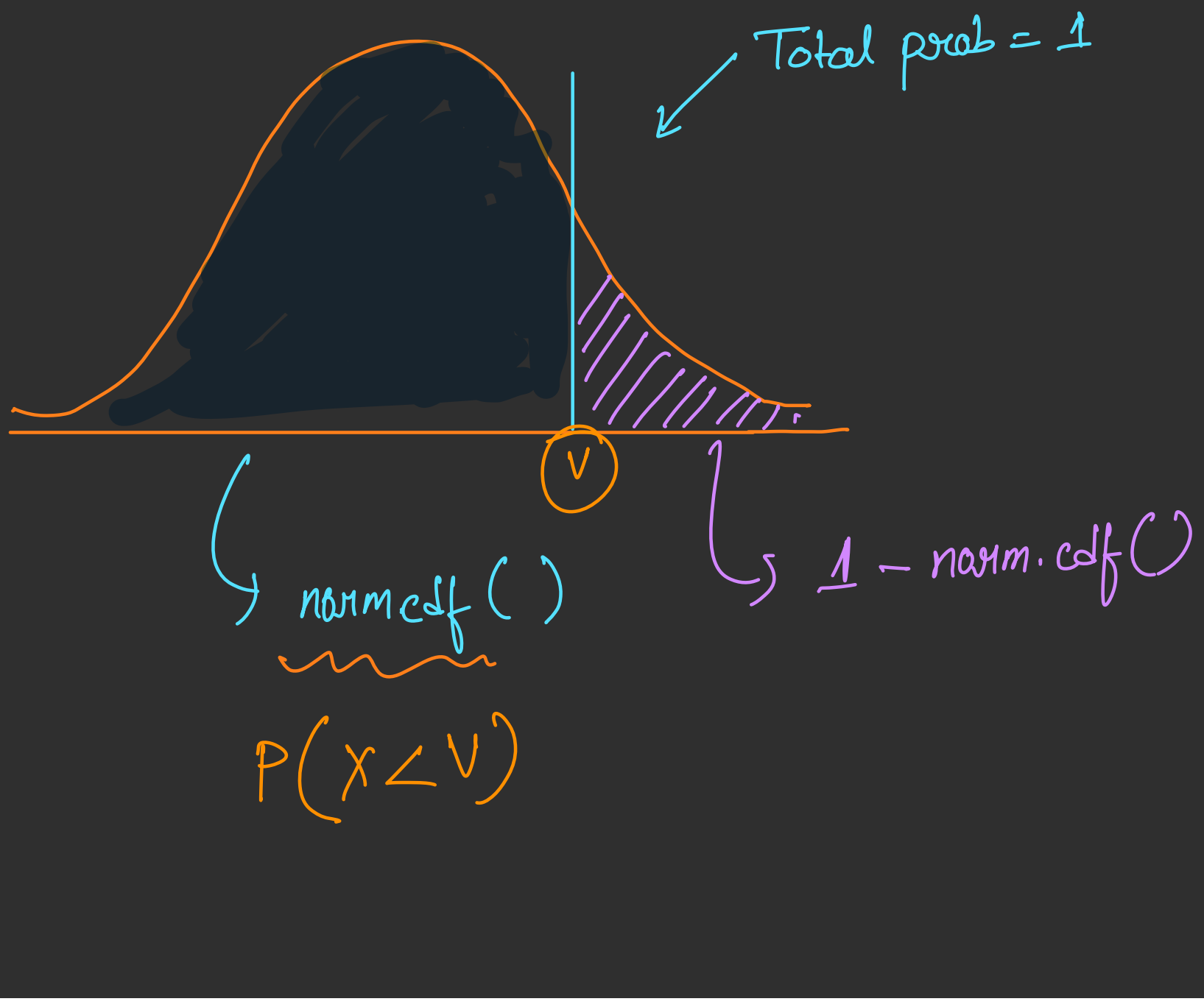
↳ z score

pdf $\xrightarrow{\text{cdf}}$

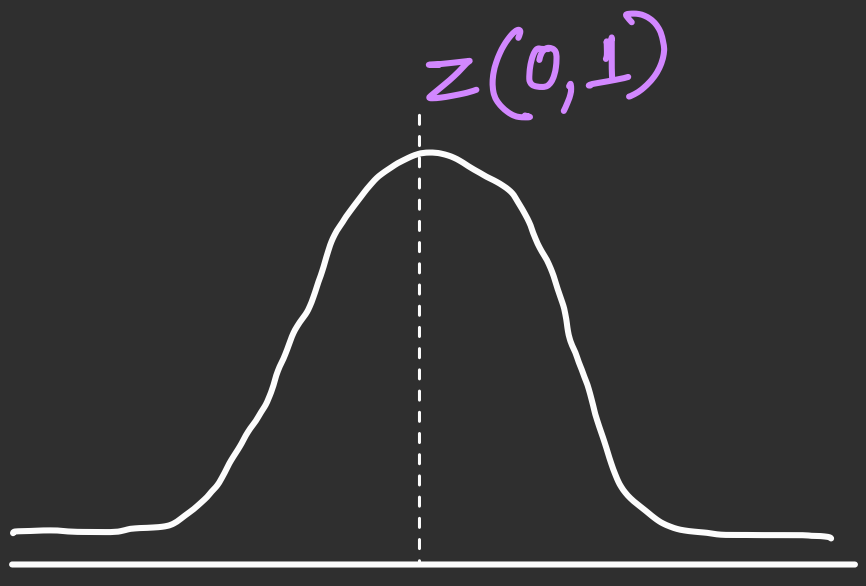
{ z score $\longrightarrow$ percentile

percentile $\longrightarrow$ zscore } ppf

$$z = \frac{x - \mu}{\sigma}$$


Total prob = 1
v
norm.cdf( )
1 - norm.cdf( )
P(X<v)

Standard Normal Dist



"Special Case of Normal Dist"

{ "Standardize Normal"
  { $\mu = 0$
  { $\sigma = 1$

{ $\mu = 0$
{ $\sigma = 1$

b/

"7 → 7:15 → 7:15 → 7:30"

3+12 = (5)

(0 < x < 3) + (15 < x < 10) =>

3/30 + 3/30

6/10 = (1/5)